

Electrical Machine- II (BT EE403A)



हरियाणा केंद्रीय विश्वविद्यालय
CENTRAL UNIVERSITY OF HARYANA
NAAC ACCREDITED 'A' GRADE UNIVERSITY

Unit-IV

Synchronous Machine

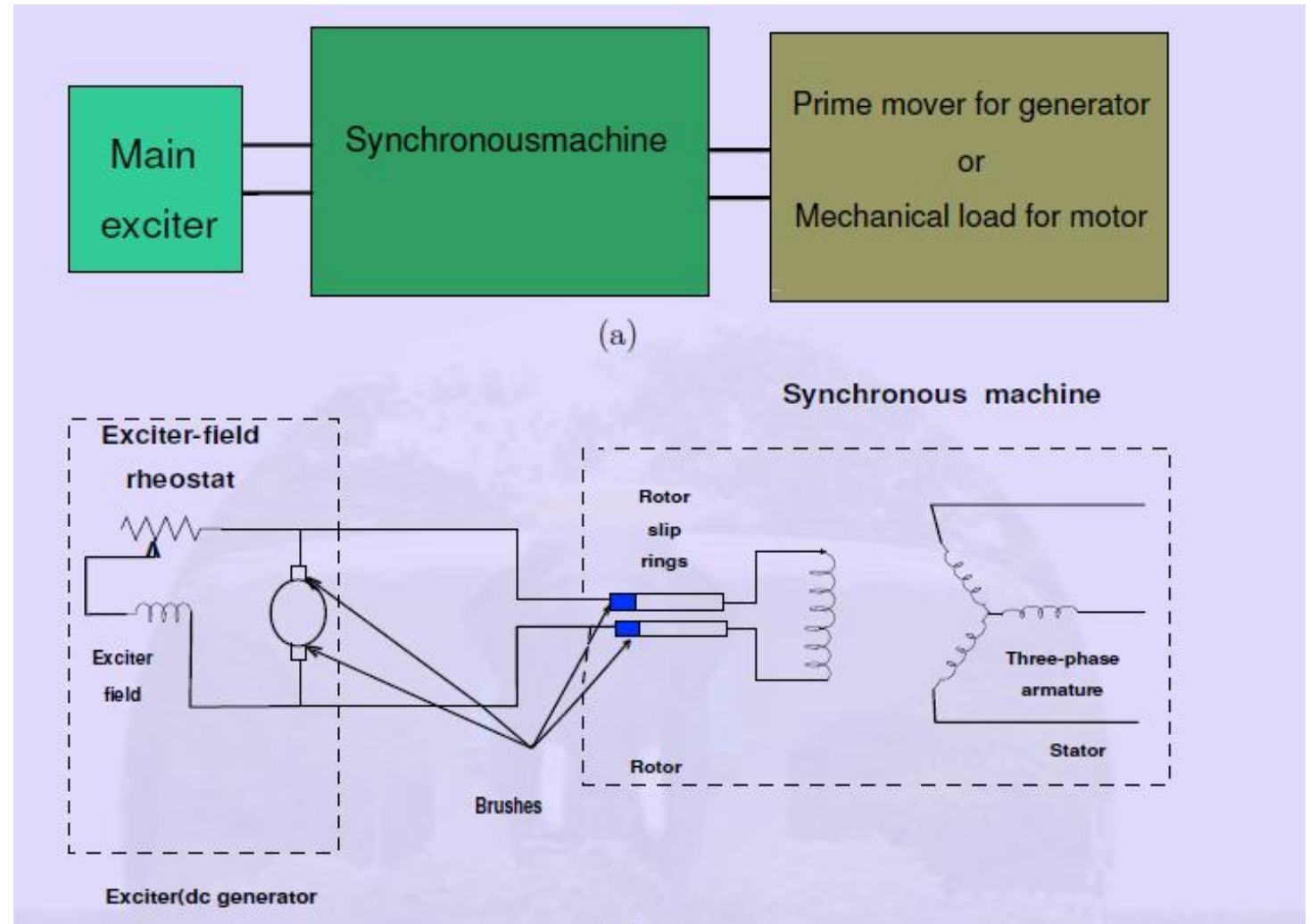
- ❖ Excitation Systems for Synchronous Machines
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Excitation Systems for Synchronous Machines

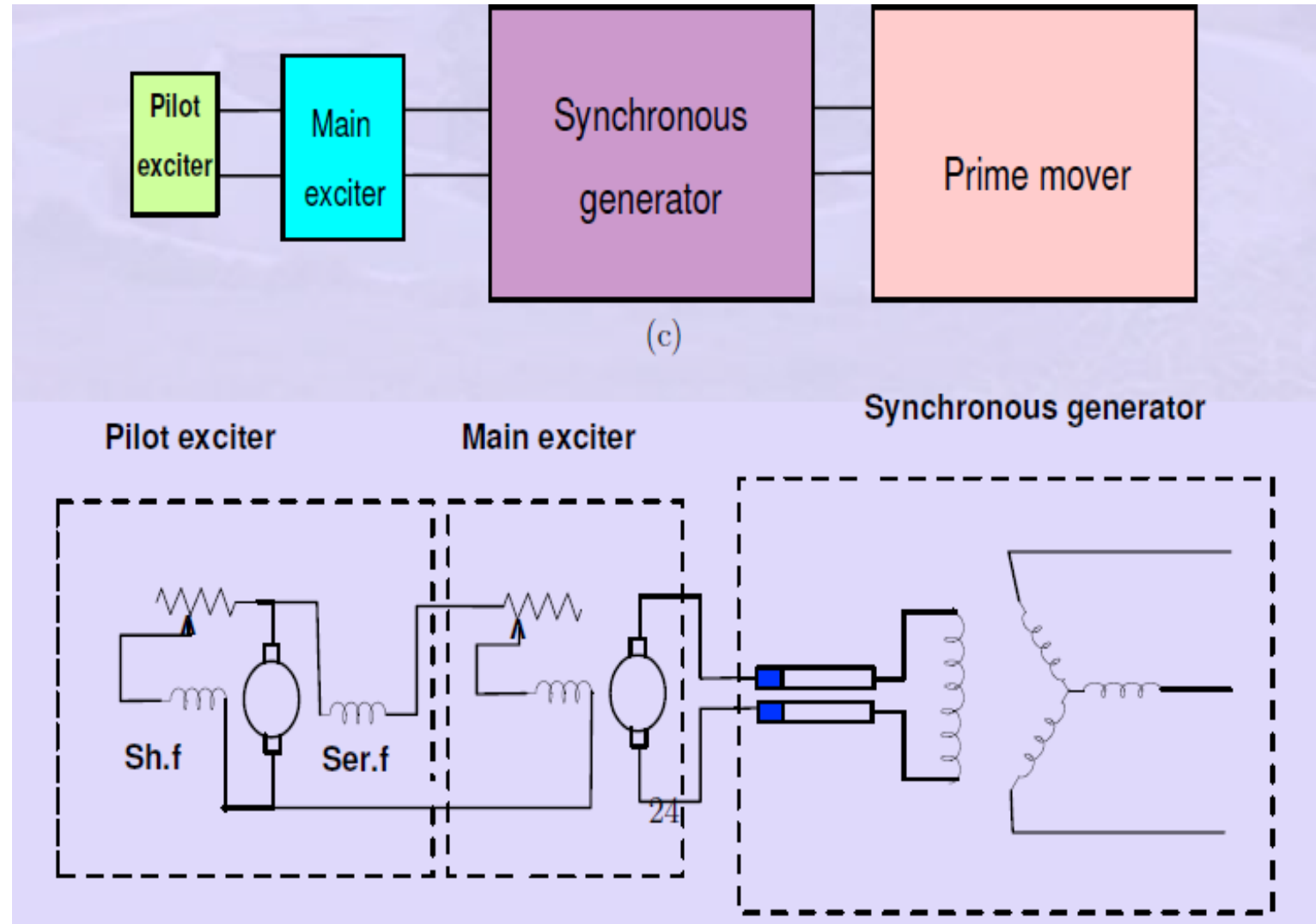
➤A number of arrangements for supplying direct current to the fields of synchronous machines have come into use. Adjustments in the field current may be automatic or manual depending upon the complexity and the requirements of the power system to which the generator is connected.

➤Excitation systems are usually 125 V up to ratings of 50kW with higher voltages for the larger ratings. The usual source of power is a direct-connected exciter, motor- generator set, rectifier, or battery.



Excitation Systems for Synchronous Machines

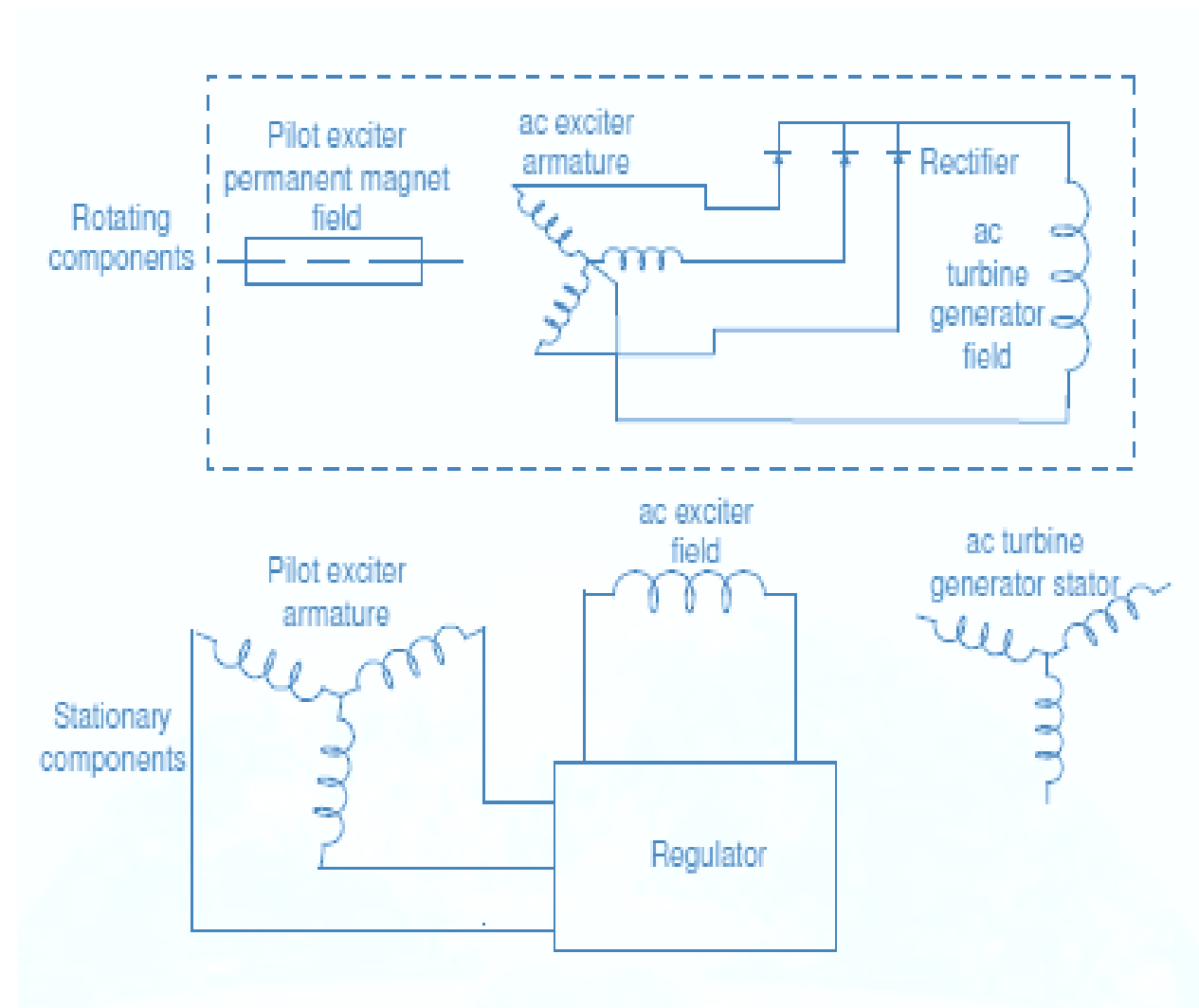
➤The output of the exciter (i.e., the field current of the synchronous machine) is varied by adjusting the exciter field rheostat. A somewhat more complex system that makes use of a pilot exciter, a compound dc generator also mounted on the generator shaft, which in turn excites the field of the main exciter, is shown in fig.



Brushless Excitation System

➤To make the excitation of a generator completely independent of any external power source, a small pilot exciter is often added to the circuit.

➤The pilot exciter is an AC generator with a permanent magnet mounted on the rotor shaft and a 3-phase winding on the stator producing the power for the field circuit of the exciter.



Action of the Synchronous Machine

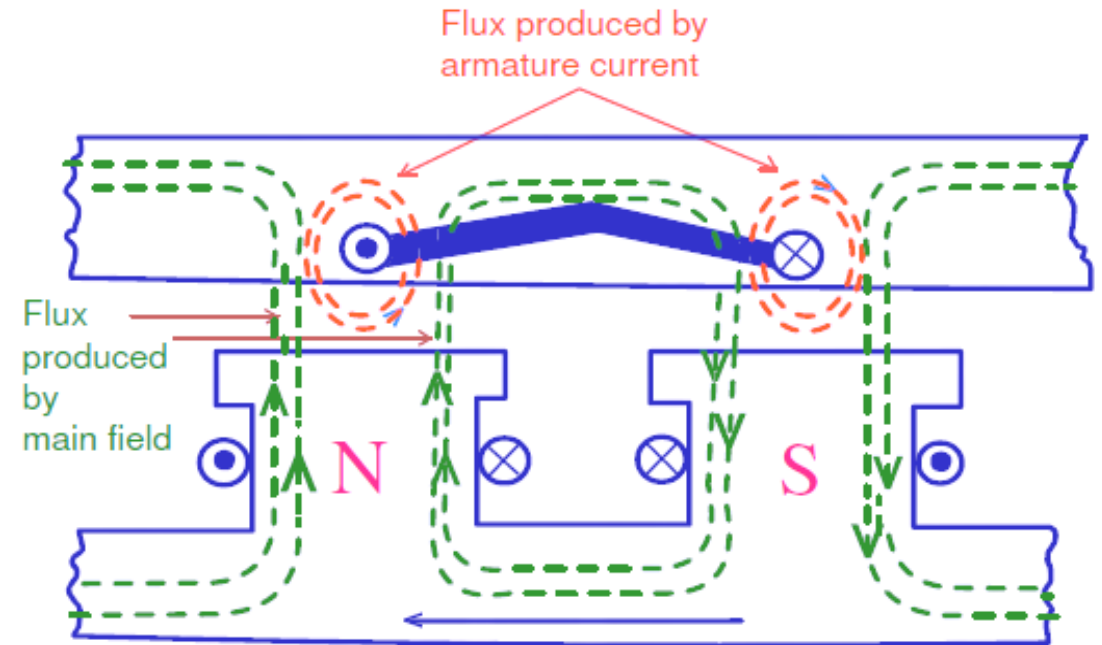
- Just like the DC generator, the behavior of a Synchronous generator connected to an external load is not the same as at no-load.
- In order to understand the action of the Synchronous machine when it is loaded, let us take a look at the flux distributions in the machine when the armature also carries a current.
- Unlike in the DC machine here the current peak and the emf peak will not occur in the same coil due to the effect of the pf of the load. In other words the current and the induced emf will be at their peaks in the same coil only for upf loads.
- For zpf (lagging) loads, the current reaches its peak in a coil which falls behind that coil wherein the induced emf is at its peak by nearly 90 electrical degrees or half a pole-pitch.
- For zpf (leading) loads, the current reaches its peak in a coil which is ahead of that coil wherein the induced emf is at its peak by nearly 90 electrical degrees or half a pole-pitch..

Armature Reaction

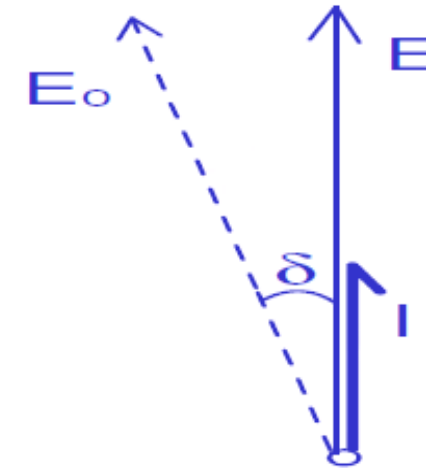
Stretched out synchronous generator



when the emf induced in the stator coils is the maximum. The coil links no resultant flux but is in the position of greatest rate of change of flux. The coil position shown is also that for maximum current when the current is in phase with the voltage: i.e for a pure resistive load. The current in the coil has no effect on the total flux per pole, but causes a strengthening on one side and a weakening on the other side of the pole shoes. Thus the armature conductors and themselves in the circumstances illustrated in Fig, and a torque is produced by the interaction of the main flux m with the current in the conductors.



(a) The effect of armature current while supplying a pure resistance load



(b) Phasor diagram

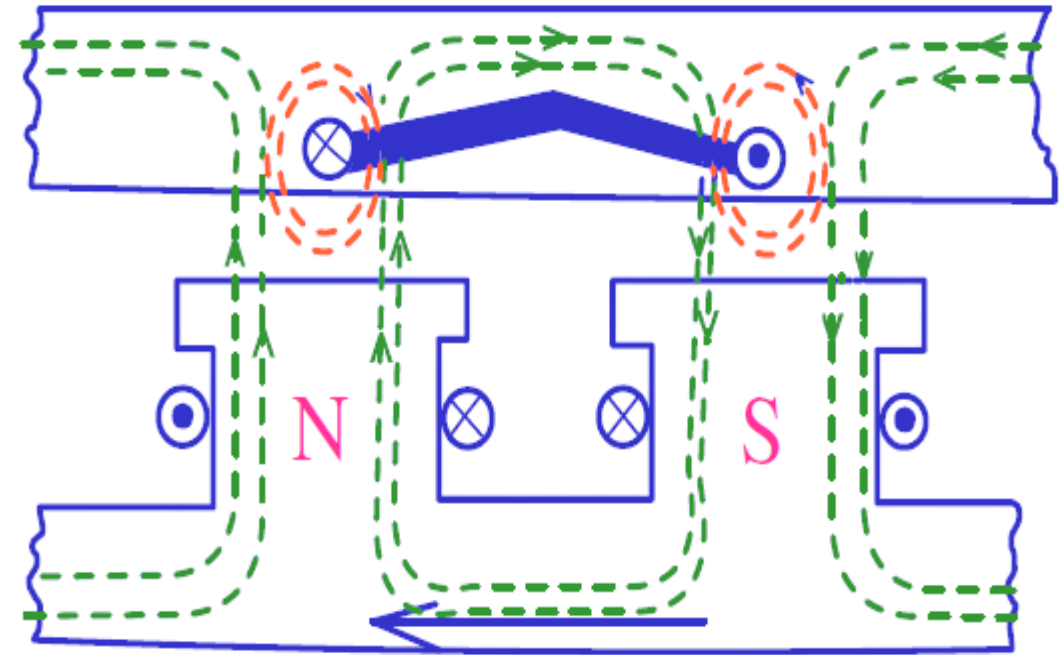
Stretched out synchronous generator

- The torque thus produced is seen to be opposed to the direction of motion of the rotor - the force on the conductors is such as to push them to the left and by reaction to push the rotor to the right (as the armature coils are stationary). The rotor is rotated by a prime mover against this reaction, so that the electrical power, the product EI , is produced by virtue of the supply of a corresponding mechanical power.
- Thus it is evident from the distortion of the main flux distribution that electrical energy is converted from mechanical energy and the machine operates as a generator. An unidirectional torque is maintained as the stator conductors cut N-Pole and S-Pole fluxes alternately resulting in alternating emfs at a frequency equal to the number of pole-pairs passed per second and the currents also alternate with the emf.
- The strengthening of the resultant flux on the right of the poles and an equivalent amount of weakening on the left effectively shift the main field flux axis against the direction of rotation, so that the actual e.m.f. E induced in the armature winding is an angle behind the position E_0 that it would occupy if the flux were undistorted as shown in the adjacent phasor diagram pertaining to this condition of operation. Thus the effect of a resistive (upf) load connected to a synchronous generator is to shift the main field flux axis by what is known as cross-magnetization.

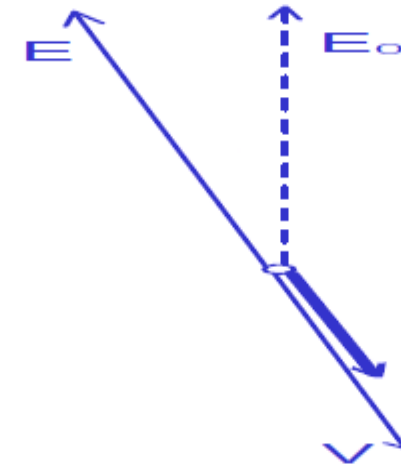
Stretched out synchronous motor



- The action of a synchronous generator operating as a motor at upf is shown in Fig.
- Just like a DC motor, a synchronous motor also requires an externally-applied voltage V in order to circulate in it a current in opposition to the induced e.m.f. E .



(a) The effect of armature current when the machine operates as a motor at u.p.f

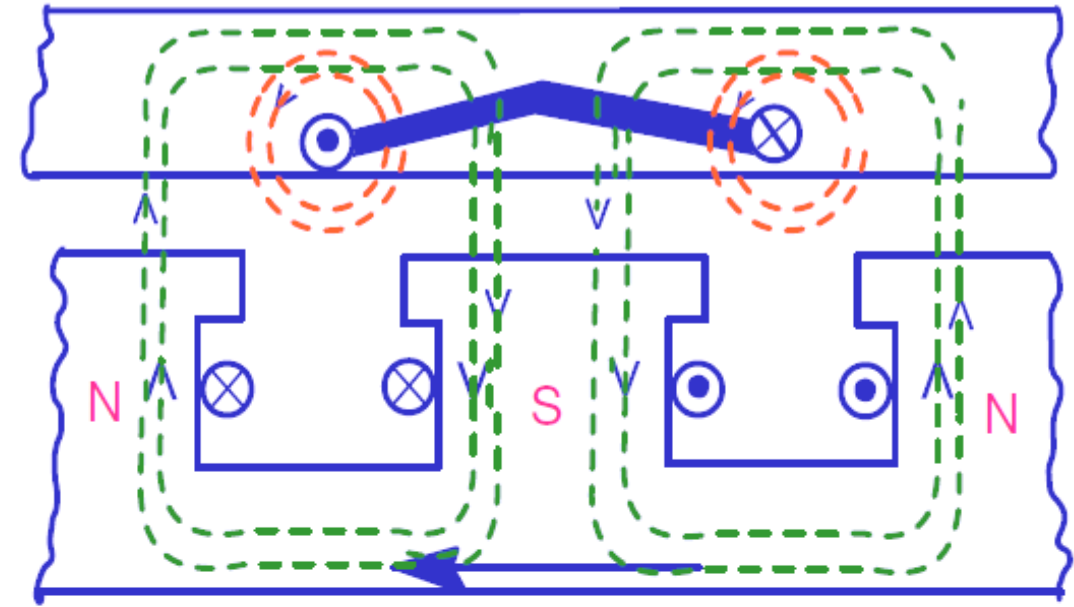


(b) Phasor diagram

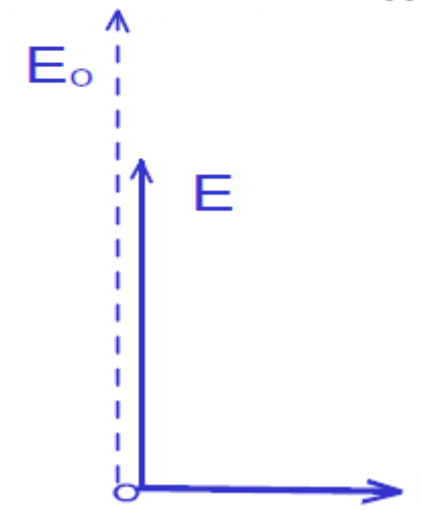
Stretched out synchronous generator With zpf (lagging) Load



As seen from this figure it is obvious that the ampere-turns of the stator coils are now in direct opposition to those on the pole, thereby reducing the total flux and e.m.f. Since the stator and rotor ampere-turns act in the same direction, there is no flux-distortion, no torque, and hence no additional mechanical power. This circumstance is in accordance with the fact that there is also no electrical power output as E and I are in phase quadrature. The phasor E_o represents the e.m.f. with no demagnetizing armature current, emphasizing the reduction in e.m.f. due to the reduced flux.



(a) The effect of armature current when it supplies a zpf(lagging) load

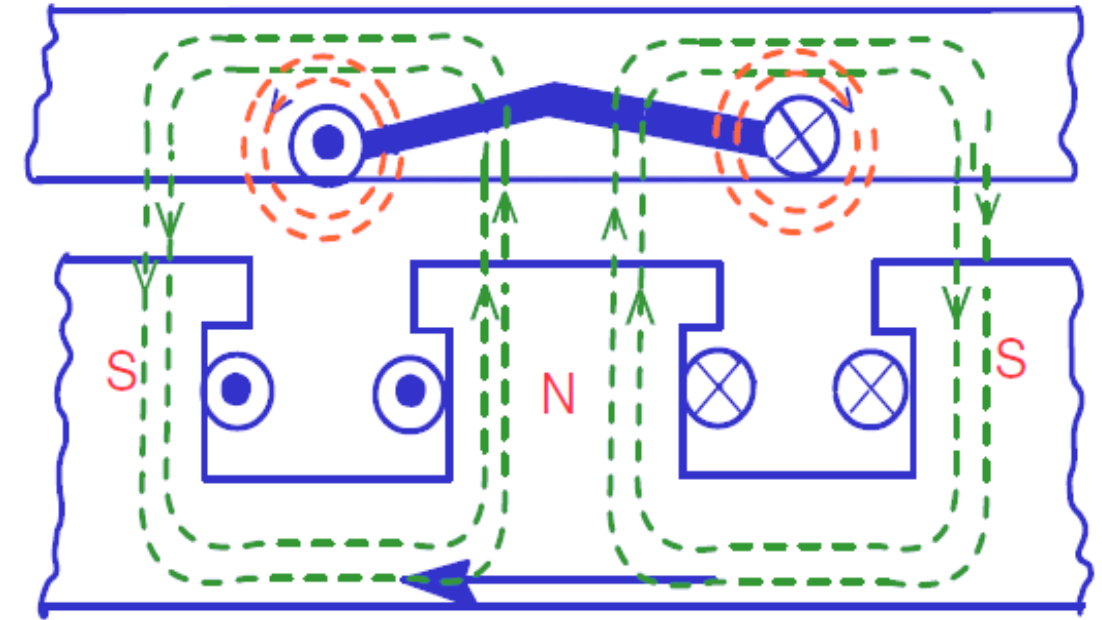


(b) Phasor diagram

Stretched out synchronous generator With zpf (leading) Load

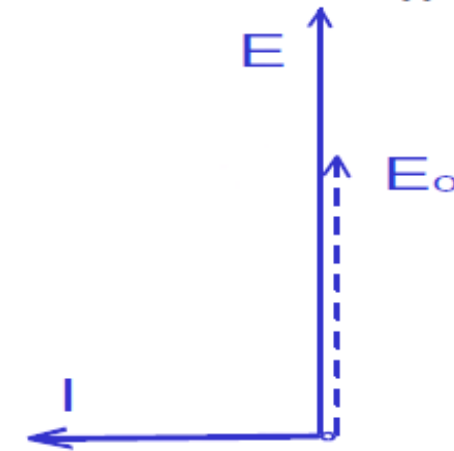


➤ when this generator is connected to a purely capacitive load i.e the current I in the coil leads the emf E by 90 electrical degrees, the conditions are such that the armature AT and the field AT will be assisting each other as shown in Fig



(a) The effect of armature current when it supplies a zpf(leading) load

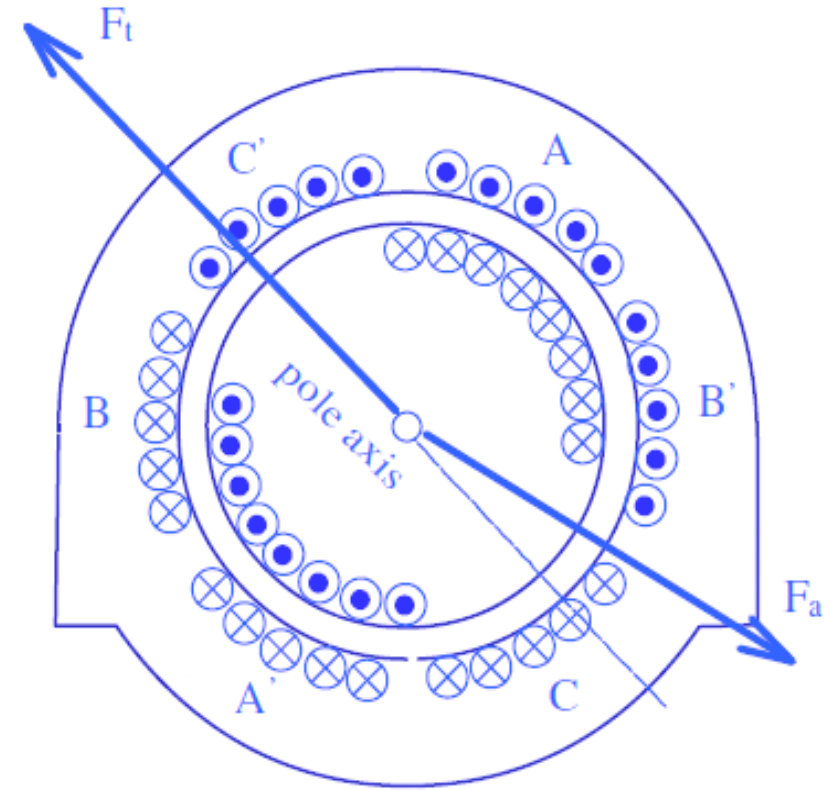
➤ When the generator supplies a load at any other power factor intermediate between unity and zero, a combination of cross- and direct-magnetization is produced on the magnetic circuit by the armature current. The cross-magnetization is distorting and torque-producing ; the direct-magnetization decreases (for lagging currents) or increases (for leading currents) the ampere-turns acting on the magnetic circuit, effecting the main flux and the e.m.f. accordingly.



(b) Phasor diagram

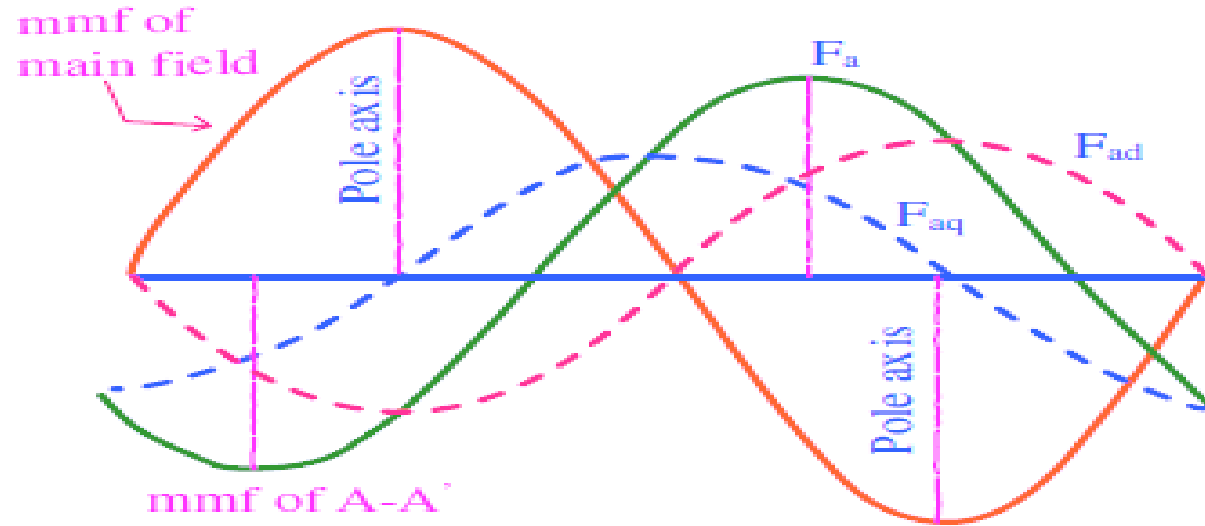
Behaviour of a loaded synchronous generator

A synchronous machine driven as a generator produces e.m.f.'s in its armature windings at a frequency $f = np$. These e.m.f.'s when applied to normal circuits produce currents of the same frequency. Depending on the p.f of the load, field distortion is produced, generating a mechanical torque and demanding an input of mechanical energy to satisfy the electrical output. As the stator currents change direction in the same time as they come from one magnetic polarity to the next, the torque is unidirectional. The torque of individual phases is pulsating just like in a single-phase induction machine - but the torque of a three-phase machine is constant for balanced loads.



Synchronous generator supplying a lagging pf load

➤ The three-phase armature winding produce a reaction Field having a sinusoidally-distributed fundamental component and an axis coincident, for the instant considered, with that of one phase such as A-A'.



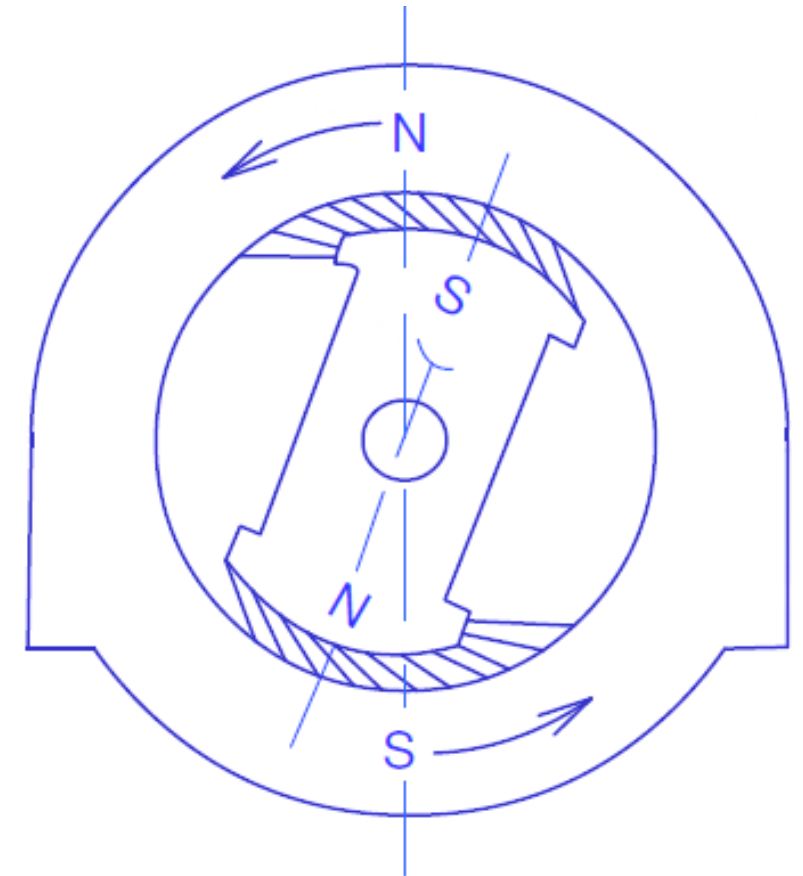
Sinusoidal distribution of the components of armature reaction in a synchronous generator

➤ The rotor windings, energized by direct current, give also an approximately sinusoidal rotor m.m.f. distribution. The machine is shown in operation as a generator supplying a lagging current.

➤ The relation of the armature reaction m.m.f. F_a to the field m.m.f. F_t is shown in Fig. The F_a sine wave is resolved into the components F_{aq} corresponding to the cross-component and F_{ad} corresponding to the direct-component, which in this case demagnetizes. F_{ad} acts in direct opposition to F_t and reduces the effective m.m.f. acting round the normal magnetic circuit. F_{aq} shifts the axis of the resultant m.m.f. (and flux) backward against the direction of rotation of the field system.

Behaviour of a loaded synchronous motor

- When a synchronous machine operates as a motor with a mechanical load on its shaft, it draws an alternating current which interacts with the main flux to produce a driving torque. The torque remains unidirectional only if the rotor moves one pole-pitch per half-cycle; i.e. it can run only at the synchronous speed.
- It is to be noted that the magnetic field shown in Fig. It is only diagrammatic and for better understanding of the action of the synchronous machine - the Flux lines may be considered as elastic bands which will be stretched by application of the mechanical load on the shaft. Actually the flux lines will enter or leave the stator and rotor surfaces nearly normally, on account of the high permeability of these members. In a salient-pole machine the torque is developed chiefly on the sides of the poles and on the sides of the teeth in a non-salient-pole machine.



Elementary synchronous motor action- Attraction of the unlike poles keep the rotor locked to the rotating field produced in the stator